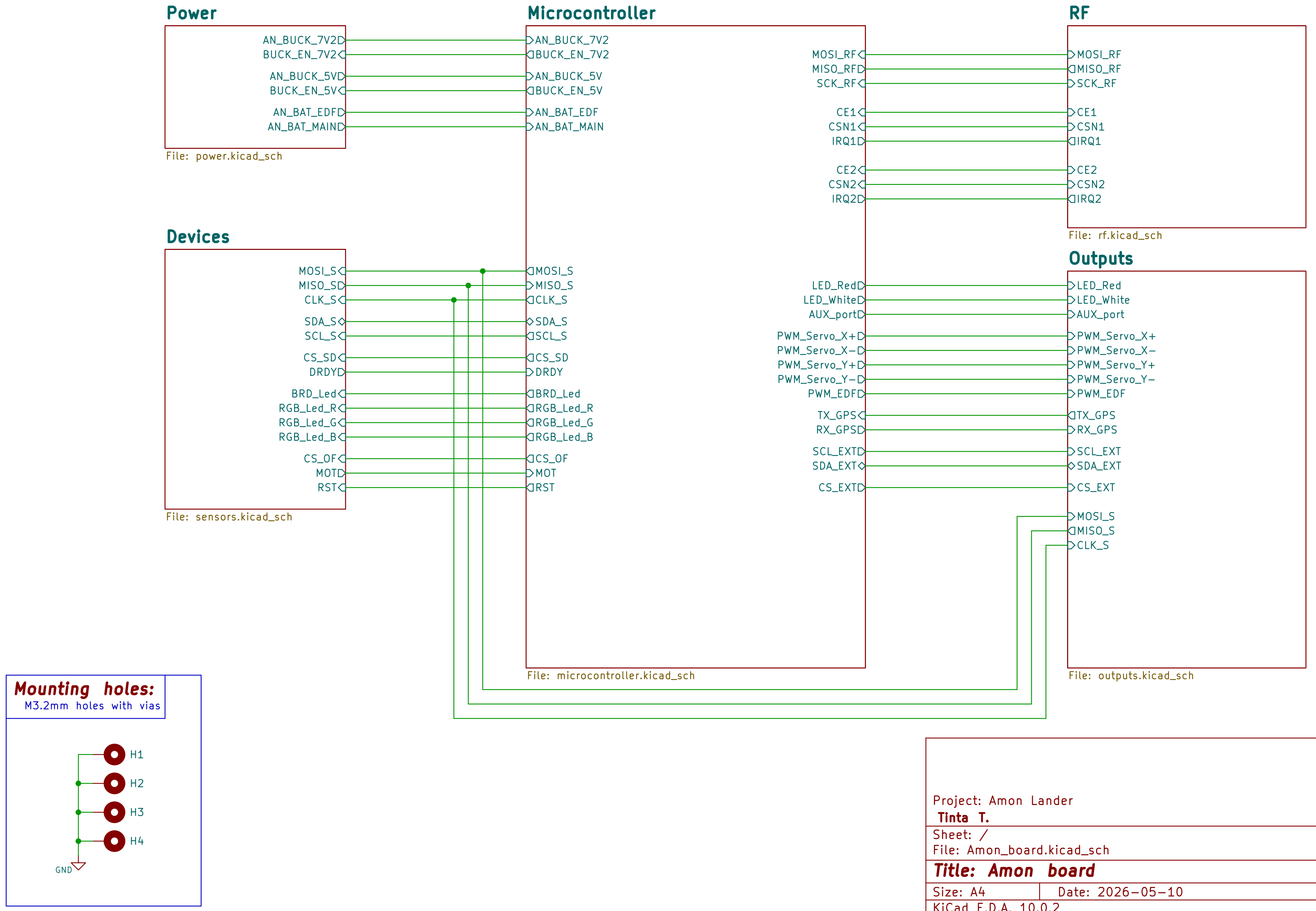
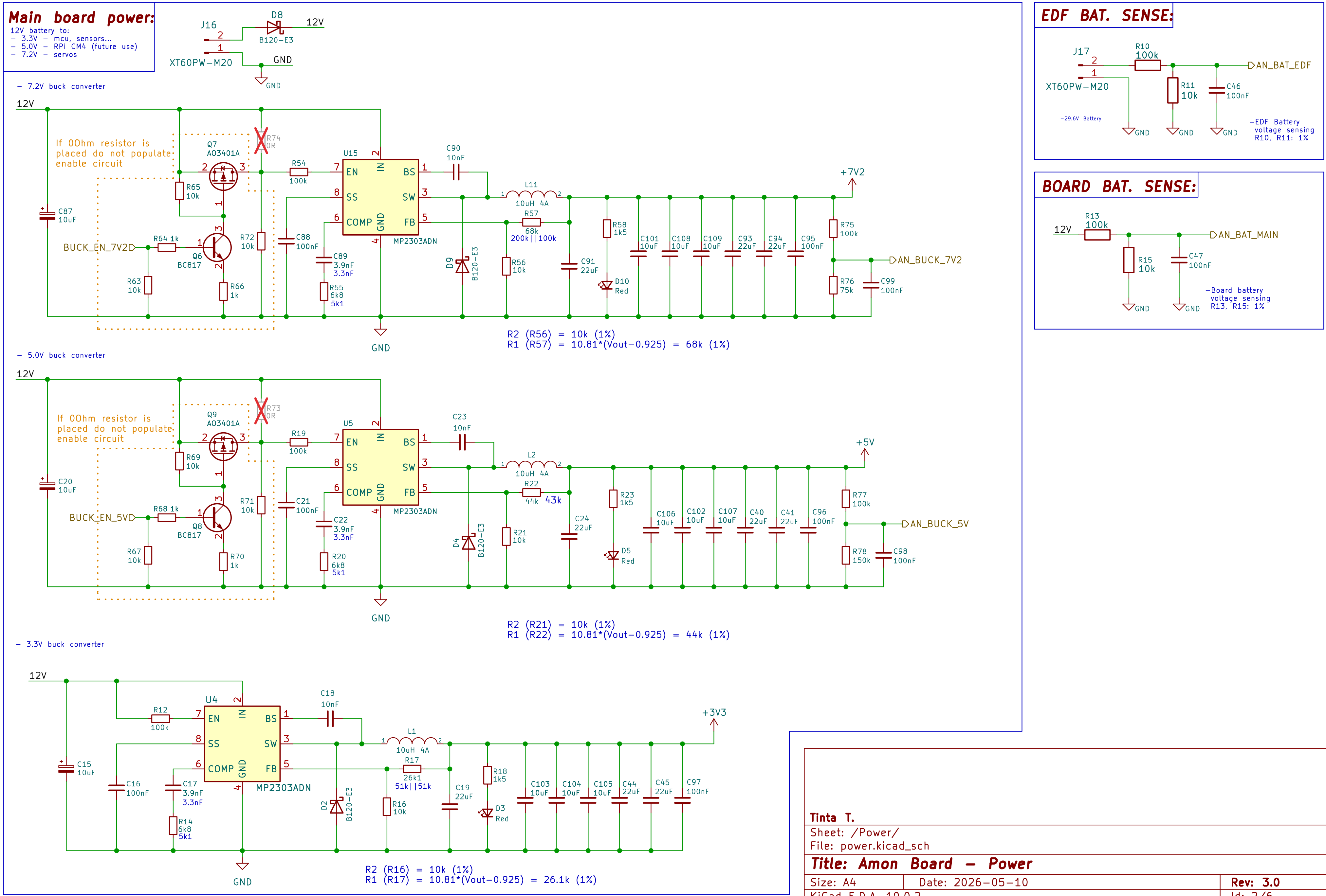
AMON Board v3



Project: Amon Lander
Tinta T.
Sheet: /
File: Amon_board.kicad_sch
Title: Amon board

Size: A4	Date: 2026-05-10	**Rev: 3.0**
KiCad E.D.A. 10.0.2		Id: 1/6



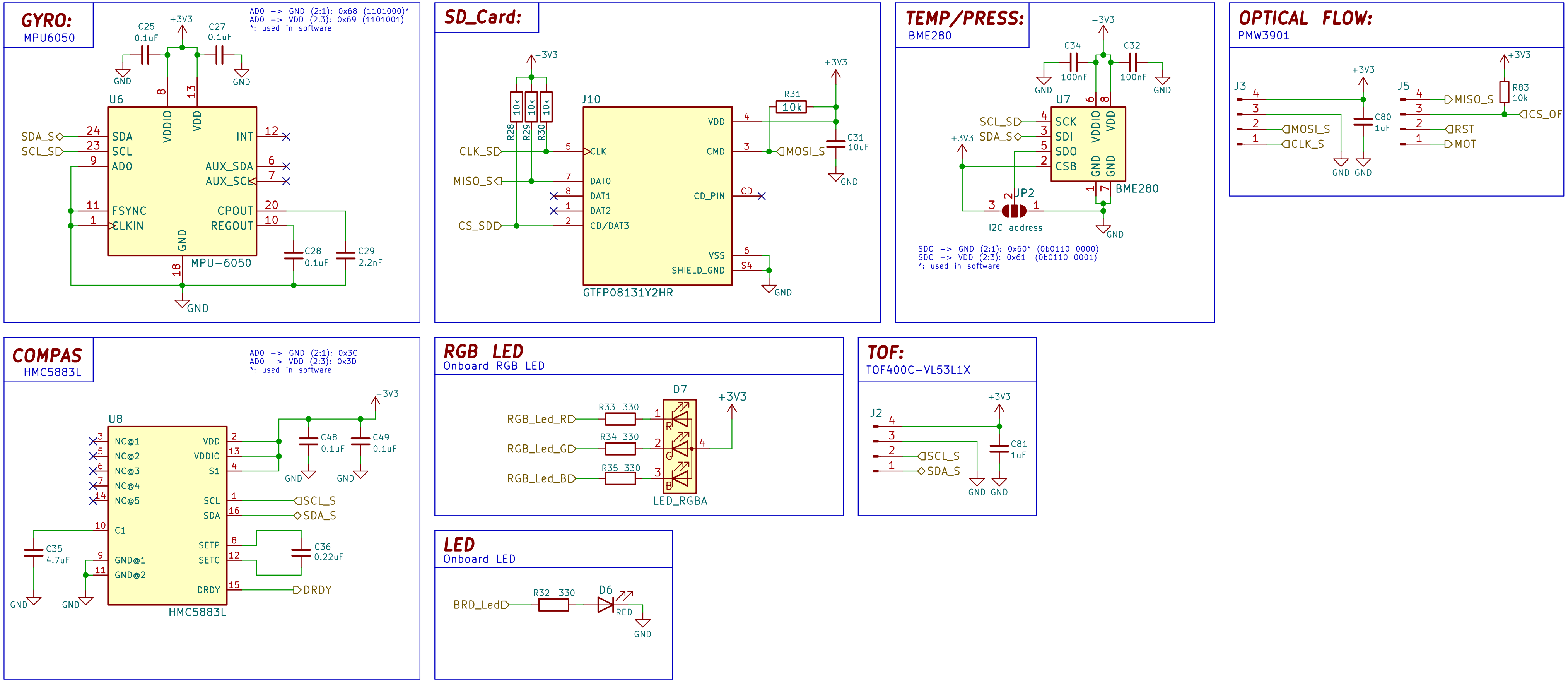
Tinta T.

Sheet: /Power/

File: power.kicad_sch

Title: Amon Board – Power

Size: A4	Date: 2026-05-10	Rev: 3.0
KiCad E.D.A. 10.0.2		Id: 2/6

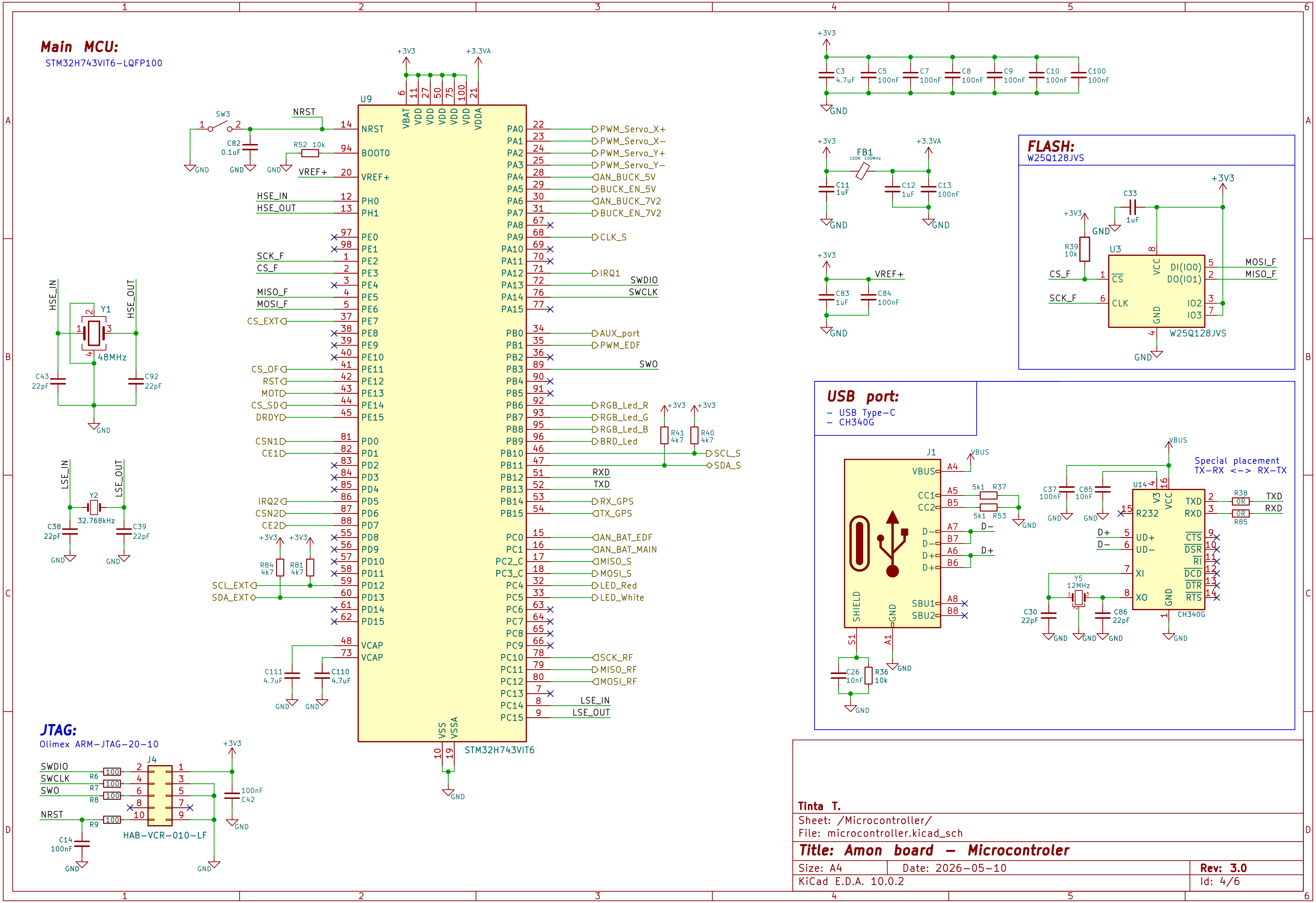


Tinta T.

Sheet: /Devices/
File: sensors.kicad_sch

Title: Amon board - Devices

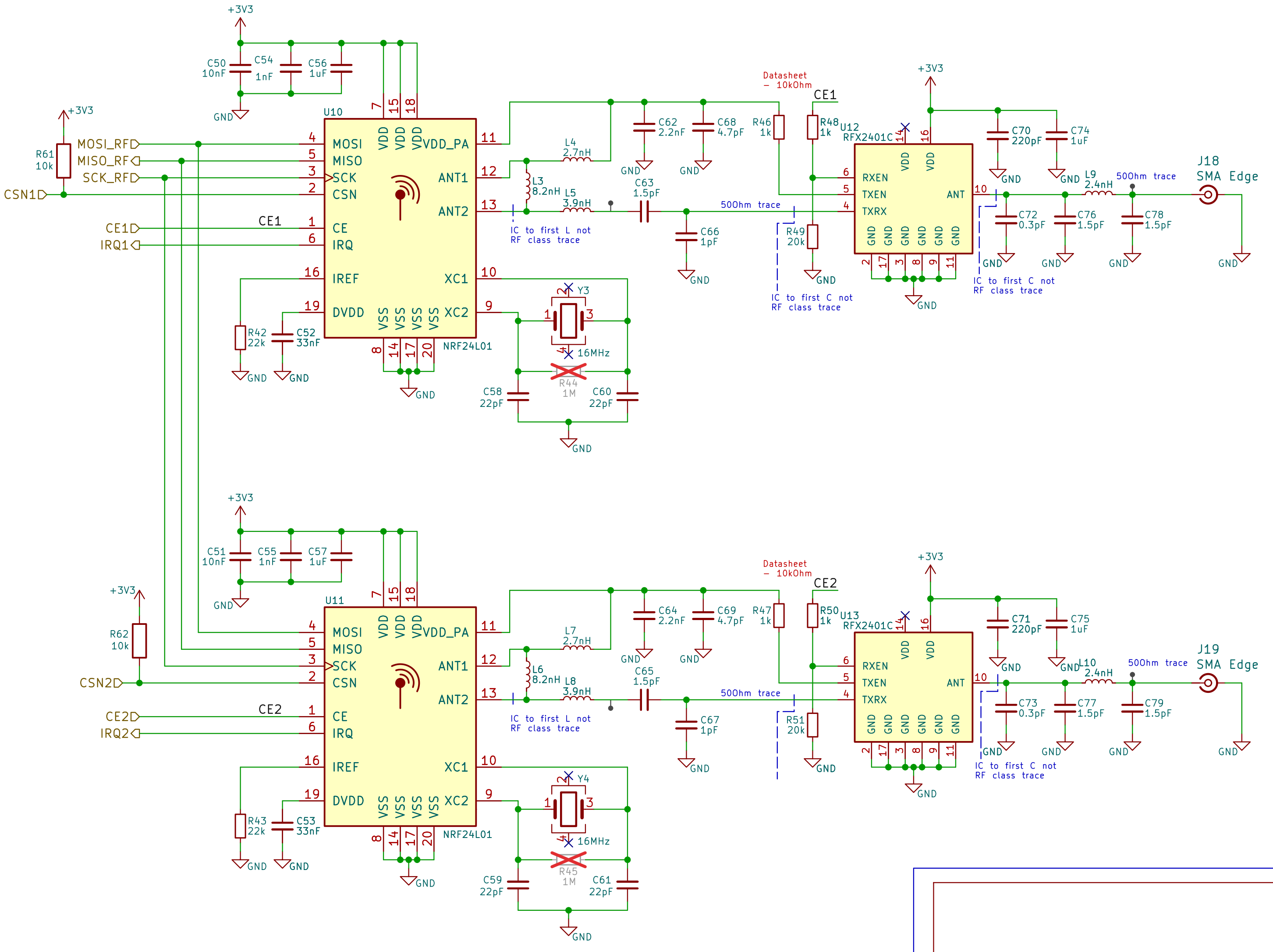
Size: A4	Date: 2026-05-10	**Rev: 3.0**
KiCad E.D.A. 10.0.2		Id: 3/6


Main MCU:
STM32H743VIT6-LQFP100
U9
STM32H743VIT6
NRST
BOOT0
VREF+
HSE_IN
HSE_OUT
SW3
C82 0.1uF
R52 10k
+3V3
+3.3VA
VBAT VDD VDD VDD VDD VDD VDDA
VSS VSSA
VCAP
C111 4.7uF
C110 4.7uF
PWM_Servo_X+
PWM_Servo_X-
PWM_Servo_Y+
PWM_Servo_Y-
AN_BUCK_5V
BUCK_EN_5V
AN_BUCK_7V2
BUCK_EN_7V2
CLK_S
IRQ1
SWDIO
SWCLK
AUX_port
PWM_EDF
SWO
RGB_Led_R
RGB_Led_G
RGB_Led_B
BRD_Led
SCL_S
SDA_S
R41 4k7
R40 4k7
RXD
TXD
RX_GPS
TX_GPS
AN_BAT_EDF
AN_BAT_MAIN
MISO_S
MOSI_S
LED_Red
LED_White
SCK_RF
MISO_RF
MOSI_RF
LSE_IN
LSE_OUT
SCK_F
CS_F
MISO_F
MOSI_F
CS_EXT
CS_OF
RST
MOT
CS_SD
DRDY
CSN1
CE1
IRQ2
CSN2
CE2
R84 4k7
R81 4k7
SCL_EXT
SDA_EXT
Y1
48MHz
C43 22pF
C92 22pF
Y2
32.768kHz
C38 22pF
C39 22pF
GND
JTAG:
Olimex ARM-JTAG-20-10
J4
HAB-VCR-010-LF
SWDIO
SWCLK
SWO
NRST
R6 100
R7 100
R8 100
R9 100
C14 100nF
C42 100nF
C3 4.7uF
C5 100nF
C7 100nF
C8 100nF
C9 100nF
C10 100nF
C100 100nF
FB1
100R 100MHz
C11 1uF
C12 1uF
C13 100nF
C83 1uF
C84 100nF
FLASH:
W25Q128JVS
U3
C33 1uF
R39 10k
CS
CLK
VCC
DI(IO0)
DO(IO1)
IO2
IO3
W25Q128JVS
USB port:
- USB Type-C
- CH340G
J1
VBUS
CC1
CC2
D-
D+
SBU1
SBU2
SHIELD
R37 5k1
R53 5k1
C26 10nF
R36 10k
U14
CH340G
C37 100nF
C85 10nF
R232
UD+
UD-
XI
XO
TXD
RXD
CTS
DSR
RI
DCD
DTR
RTS
R38 0R
R85 0R
Y5 12MHz
C30 22pF
C86 22pF
Special placement
TX-RX <-> RX-TX
Tinta T.
Sheet: /Microcontroller/
File: microcontroller.kicad_sch
Title: Amon board - Microcontroler
Size: A4
Date: 2026-05-10
KiCad E.D.A. 10.0.2
Rev: 3.0
Id: 4/6

RADIOS:

- Dual 2.4GHz radios
- Nordic nRF24L01
- Skyworks RFX2401C 2.4GHz(LNA & PA)
- ALL NC pins/pads connect to GND!

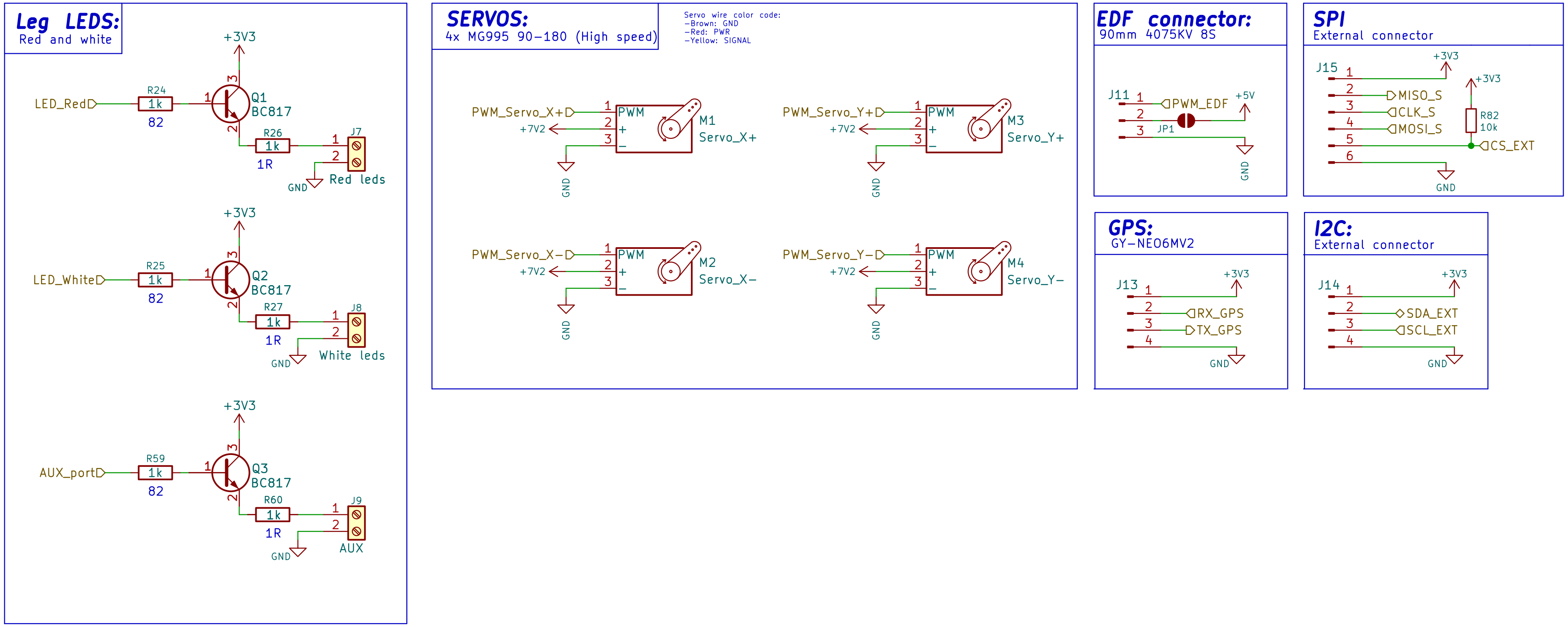
- Via stiching
  FR4 (epsilon_relative): ~4.2 - 4.8
  f = 2.4GHz = 2.4*10^9 Hz
  v = c / sqrt(epsilon_relative) = (3*10^8) / (sqrt(4.5) = ~1.42*10^8
  lambda = v / f = (1.42*10^8) / 2.4*10^9 = 0.059
  lambda / 10 = 0.059 / 10 = 0.0059 = ~5.9mm



Tinta T.
Sheet: /RF/
File: rf.kicad_sch

Title: Amon board - RF

Size: A4	Date: 2026-05-10	Rev: 3.0
KiCad E.D.A. 10.0.2		Id: 5/6



Tinta T.

Sheet: /Outputs/
File: outputs.kicad_sch

Title: Amon board - Outputs

Size: A4	Date: 2026-05-10	**Rev: 3.0**
KiCad E.D.A. 10.0.2		Id: 6/6